

Assignment 3 -

Linear algebra and python programming.

part : 1 linear algebra fundamentals

Q1 → Matrix operations
Given :

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

we have to calculate $A \times B$.

Step-by-step multiplication

$$A \times B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

First element :

$$(1 \times 5) + (2 \times 7) = 5 + 14 = 19$$

$$(1 \times 6) + (2 \times 8) = 6 + 16 = 22$$

$$(3 \times 5) + (4 \times 7) = 15 + 28 = 43$$

$$(3 \times 6) + (4 \times 8) = 18 + 32 = 50$$

Therefore,

$$A \times B = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Q2)
→

Determinant and inverse
Crises

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

For a 2×2 matrix :

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

determinant is :

$$|A| = ad - bc$$

Therefore,

$$\begin{aligned} |A| &= (1 \times 4) - (2 \times 3) \\ &= 4 - 6 = -2 \end{aligned}$$

So,

$$\boxed{|A| = -2}$$

Since determinant is not zero, the inverse exist

formula :

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Therefore,

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

Hence,

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$$

Q3 → System of Linear Equations

Given:

$$2x + y = 5$$

$$-x + 3y = 8$$

using Substitution

from the first equation:

$$y = 5 - 2x$$

Substitute this in the second equation:

$$-x + 3(5 - 2x) = 8$$

$$-x + 15 - 6x = 8$$

$$-7x + 15 = 8$$

$$-7x = -7$$

$$x = 1$$

Now,

$$y = 5 - 2(1)$$

$$y = 3$$

Final Answer

$$x = 1, \quad y = 3$$

Q4 Eigenvalues and Eigenvectors

An eigenvalue is a scalar value λ such that for a square matrix A , there exists a non-zero vector v satisfying

$$Av = \lambda v$$

Here, λ is called the eigenvalue.

Eigenvector

An eigenvector is a non-zero vector that does not change its direction when a linear transformation represented by matrix A is applied to it.

Mathematically :

$$Av = \lambda v$$

where :

- A = square matrix
- v = eigenvector
- λ = Eigenvalue